SETTING UP A TELESCOPE

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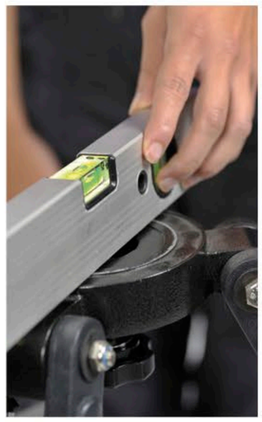
THE SKY IS CLEAR, the forecast is good, and your first night of observing lies ahead. However, there is a steep learning curve to negotiate before you can start to see the sky's wonders. Even relatively simple telescopes can magnify objects many dozens of times, so locating apparently obvious bright objects can be surprisingly difficult. The secret to successful observations is to get your bearings before you begin. It may seem obvious, but make sure you know where north and south are—even go-to telescopes may need you to point them in the right direction initially.

MOUNTING A TELESCOPE

After buying a telescope, it is important to take time to set up its optics, tripod, and mount properly. Careful setup will leave you with a well-aligned and balanced telescope that is a joy to use and that will require minimum tweaking during those precious observing hours. Each telescope is different, so be sure to read the instructions provided before you start or, better still, ask an experienced astronomer to take you through the basics. Below is a brief and general guide to the main points of setting up a typical amateur telescope—a reflector on a motorized equatorial mount. You will probably want to leave your telescope partly set up between observing sessions, so some of the steps will only need to be carried out the first time you use it.

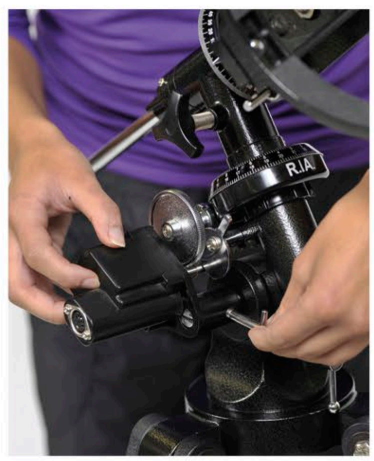


1 LEVEL TRIPOD Set up your tripod on solid, level ground. Use a level to check that the top plate of the tripod is horizontal and adjust the tripod legs as necessary.

2 ADJUST LEGS Avoid extending the sections of the tripod legs to their full extent, because this makes the platform less stable and gives you no latitude for fine adjustment of height later. Double-check that the locks on the legs are secure.

3 PLACE MOUNT Gently position the mount onto the tripod, ensuring that the protrusion on the mount slots into the hole on the tripod.

4 SECURE MOUNT Tighten the mounting screw from beneath the tripod head, making sure it is completely secure.

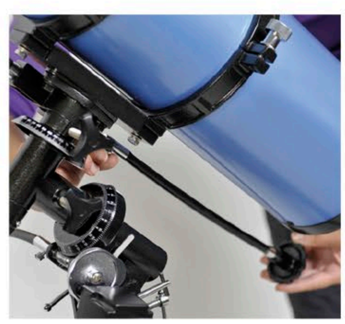


5 ATTACH MOTOR DRIVE Attach the motor drive to the mount and ensure that the gears of the motor are correctly engaged with those on the mount.

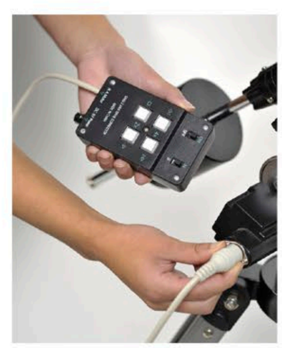
6 ALIGN NORTH If using an equatorial mount, check that the right ascension axis (the long part of the central "T" of the mount) is pointing roughly toward the north (or south) celestial pole, depending on your hemisphere.

7 ADD COUNTERWEIGHTS Slot the counterweights onto the counterweight shaft and use the nut to secure the weights in position. There is usually a safety screw at the end of the shaft that stops the counterweights from sliding off should the main nut fail. Be sure to replace this safety screw after positioning the weights.

8 MOUNT THE TELESCOPE Once the mount is on the tripod, you can mount the telescope tube. Place the tube inside the pair of circular mounting rings (called cradles) and clamp them tight around the tube using the screws.



9 ADD FINE ADJUSTMENT CABLES Screw in the fine adjustment cables—these will allow you to make small changes to the right ascension and declination when observing.



10 ADD THE MOTOR UNIT Plug the drive controller into the motor unit, but do not connect it to the power supply.